SUDIPTO GAYEN

Kolkata, West Bengal, India
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SUMMARY

Backend and full-stack developer with production experience building cloud-native systems using Node.js, TypeScript, and GCP. Built **GiteaForge** — an academic Git management platform featuring Judge0 multi-language code execution, HMAC-authenticated webhook pipelines, CI/CD via GitHub Actions and GCP Cloud Run, and fault-tolerant Redis + PostgreSQL session architecture. Seeking a backend or full-stack engineering internship to contribute to high-impact teams.

EDUCATION

The Neotia University *Aug 2023 – Jun 2027*
B.Tech in Computer Science Engineering (Cybersecurity Specialization) *CGPA: 7.85 (till 5th Sem)*

Shri Ritam Vidyapith *2023*
Higher Secondary (WBBSE) *Percentage: 66%*

TECHNICAL SKILLS

Languages: C, JavaScript (ES6+), TypeScript, Python, HTML, CSS
Backend: Node.js, Express.js, RESTful APIs, Prisma ORM, JWT Auth, RBAC, HMAC Webhooks, Zod Validation
Frontend: React 19, Vite, Tailwind CSS
Database & Caching: PostgreSQL (Neon), MongoDB, Mongoose, Redis (Session Store & Caching)
Deployment & Cloud: GCP (Cloud Run, Compute Engine, Artifact Registry), AWS (EC2, S3), Docker, Vercel, Neon
DevOps & CI/CD: GitHub Actions, Docker, Prisma Migrations, ESLint, PM2
Testing & Security: Jest, Supertest, JWT, HttpOnly Cookie Auth, CSRF Protection, OWASP Top 10, VAPT
Tools & Platforms: Git, GitHub, Postman, Judge0, Gitea, Burp Suite, Kali Linux, VS Code

EXPERIENCE

AiLabs, India *Jun 2025 – Jul 2025*
Backend Development Intern *SaltLake, Kolkata*

- Developed the backend of a production-grade E-Commerce application using Node.js, Express.js, and MongoDB.
- Designed and implemented 25+ RESTful APIs across product, auth, cart, and order modules; tested across all endpoints via Postman; zero critical failures during internship period.
- Integrated JWT-based authentication and role-based access control (RBAC) for Admin and User roles.

Dataspace Academy *Jul 2024*
Cybersecurity Intern *SaltLake, Kolkata*

- Performed VAPT on web applications using Burp Suite, Kali Linux, and Nmap; documented findings against OWASP Top 10.

PROJECTS

GiteaForge — Academic Git & Project Management Platform *Live Demo*

- Architected a cloud-native CI/CD pipeline using GitHub Actions, Docker, and Google Artifact Registry to deploy containerized Express 5 services to **GCP Cloud Run** with automated Neon PostgreSQL migrations on every deployment.
- Engineered a multi-language code sandbox integrating **Judge0** execution engine on a **GCP Compute Engine VM** with Redis worker queues — supporting 8+ languages with isolated compilation, stdin simulation, and CPU/memory constraints.
- Built an **HMAC-SHA256** verified Git webhook pipeline using `timingSafeEqual` to sync Gitea push events — auto-parsing commit logs, tracking code diffs, and auto-progressing milestone deliverables in real time.
- Implemented in-platform code reviews with SHA-specific line commenting, multi-factor rubric grading (Code Quality, Innovation, Teamwork), and RBAC-protected **AWS S3 presigned URLs** with 1-hour expiration for secure artifact storage.
- Designed fault-tolerant hybrid session architecture combining Redis token verification, HttpOnly SameSite cookie rotation, and PostgreSQL fallback — eliminating Redis as a single point of failure for authenticated sessions.
- **Tech Stack:** React 19, TypeScript, Vite, Node.js, Express 5, PostgreSQL (Neon), Prisma ORM, Redis, Docker, GCP (Cloud Run, Compute Engine, Artifact Registry), AWS S3, Judge0, Gitea API, GitHub Actions, Jest, Supertest, Tailwind CSS

ShopXpress — Full-Stack E-Commerce Platform *Live Demo | GitHub*

- Full-stack MERN e-commerce platform with JWT auth, Admin/User RBAC, Admin Dashboard, and **Razorpay** payment flow with webhook verification for order confirmation.
- Responsive React UI with cart, filters, and multi-step checkout; RESTful APIs for all core e-commerce operations.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Razorpay, AWS (EC2, S3), Vercel

ACTIVITIES & CO-CURRICULAR

- **The Complete Full-Stack Web Development Bootcamp** – Udemy, Dr. Angela Yu (Dec 2025) | Certificate
- Cybersecurity Internship Certificate – Dataspace Academy (2024)
- NCC Cadet; Trekking & Rock Climbing